STC Jawwy

In [3]:
```python
import pandas as pd
import pyxlsb
import numpy as np
```

Jawwy dataset

The dataset consists of meta details about the movies and tv shows as genre. Also details about Users activities, spent duration and if watching in High definition or standard definition. You have to analyse this dataset to find top insights, findings and to solve the four tasks assigned to you.

In [ ]:
```python
```

In [ ]:
```python
# dataframe = pd.read_excel("stc_TV_Data_Set_T22.xlsx",sheet_name="Final_Dataset")
# dataframe
```

In [5]:
```python
dataframe = pd.read_csv('stc_TV_Data_Set_T222.csv')
```

In [6]:
```python
df = dataframe.copy()
```

In [9]:
```python
df
```

Out[9]:

	Column1	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	1	27/05/2017	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0	0	100 treets
1	3	21/05/2017	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0	1	Moana
2	4	10/08/2017	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
3	5	26/07/2017	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
4	7	07/07/2017	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0	1	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	1247848	22/04/2017	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0	0	Extraterrestrial
1048571	1247849	29/11/2017	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0	0	Friends
1048572	1247850	15/06/2017	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0	0	Masters of Horror
1048573	1247851	11/09/2017	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0	1	Two Lovers and a Bear
1048574	1247852	02/05/2017	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0	1	American Psycho

1048575 rows × 13 columns

In [11]:
```python
# check the data shape
dataframe.shape
```

Out[11]:
```
(1048575, 13)
```

In [13]:
```python
# check the data Type
dataframe.dtypes
```

Out[13]:
```
Column1              int64
date_               object
user_id_maped        int64
program_name        object
duration_seconds     int64
program_class       object
season               int64
episode              int64
program_desc        object
program_genre       object
series_title        object
hd                  object
original_name       object
dtype: object
```

In [15]:
```python
# display the first 5 rows
dataframe.head()
```

Out[15]:

	Column1	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	1	27/05/2017	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0	0	100 treets
1	3	21/05/2017	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0	1	Moana
2	4	10/08/2017	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
3	5	26/07/2017	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
4	7	07/07/2017	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0	1	Churchill

In [17]:
```python
dataframe = dataframe.drop(columns=['Column1']) # dropping the index column
dataframe
```

Out[17]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	27/05/2017	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0	0	100 treets
1	21/05/2017	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0	1	Moana
2	10/08/2017	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
3	26/07/2017	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
4	07/07/2017	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0	1	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	22/04/2017	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0	0	Extraterrestrial
1048571	29/11/2017	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0	0	Friends
1048572	15/06/2017	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0	0	Masters of Horror
1048573	11/09/2017	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0	1	Two Lovers and a Bear
1048574	02/05/2017	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0	1	American Psycho

1048575 rows × 12 columns

In [19]:
```python
dataframe['program_name'] = dataframe['program_name'].str.strip()  # trim spaces in movies names to avoid misspellings in input data
dataframe
```

Out[19]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	27/05/2017	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0	0	100 treets
1	21/05/2017	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0	1	Moana
2	10/08/2017	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
3	26/07/2017	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
4	07/07/2017	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0	1	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	22/04/2017	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0	0	Extraterrestrial
1048571	29/11/2017	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0	0	Friends
1048572	15/06/2017	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0	0	Masters of Horror
1048573	11/09/2017	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0	1	Two Lovers and a Bear
1048574	02/05/2017	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0	1	American Psycho

1048575 rows × 12 columns

In [21]:
```python
dataframe['date_'] = pd.to_datetime(dataframe['date_'], format='%d/%m/%Y')
dataframe
```

Out[21]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0	0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0	1	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0	1	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0	1	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2017-04-22	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0	0	Extraterrestrial
1048571	2017-11-29	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0	0	Friends
1048572	2017-06-15	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0	0	Masters of Horror
1048573	2017-09-11	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0	1	Two Lovers and a Bear
1048574	2017-05-02	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0	1	American Psycho

1048575 rows × 12 columns

In [27]:
```python
dataframe[['duration_seconds', 'season','episode','series_title','hd']] = dataframe[['duration_seconds', 'season','episode','series_title','hd']].apply(pd.to_numeric, errors='coerce')  # read numeric columns as numeric data types
dataframe
```

Out[27]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0.0	0.0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0.0	1.0	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2017-04-22	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0.0	0.0	Extraterrestrial
1048571	2017-11-29	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0.0	0.0	Friends
1048572	2017-06-15	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0.0	0.0	Masters of Horror
1048573	2017-09-11	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0.0	1.0	Two Lovers and a Bear
1048574	2017-05-02	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0.0	1.0	American Psycho

1048575 rows × 12 columns

In [29]:
```python
dataframe[['user_id_maped', 'program_name','program_class','program_desc','program_genre','original_name']] = dataframe[['user_id_maped', 'program_name','program_class','program_desc','program_genre','original_name']].astype(str) # re
dataframe
```

Out[29]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0.0	0.0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0.0	1.0	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2017-04-22	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0.0	0.0	Extraterrestrial
1048571	2017-11-29	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0.0	0.0	Friends
1048572	2017-06-15	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0.0	0.0	Masters of Horror
1048573	2017-09-11	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0.0	1.0	Two Lovers and a Bear
1048574	2017-05-02	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0.0	1.0	American Psycho

1048575 rows × 12 columns

In [31]:
```python
# display the dataset after applying data types
dataframe.head()
```

Out[31]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0.0	0.0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0.0	1.0	Churchill

In [33]:
```python
# describe the numeric values in the dataset
dataframe.describe()
```

Out[33]:

	date_	duration_seconds	season	episode	series_title	hd
count	1048575	1.048575e+06	1.048575e+06	1.048575e+06	883931.000000	904940.000000
mean	2017-10-04 00:23:20.346183936	1.230957e+03	1.342139e+00	6.157952e+00	0.008838	0.426162
min	2017-03-14 00:00:00	2.000000e+00	0.000000e+00	0.000000e+00	0.000000	0.000000
25%	2017-06-10 00:00:00	5.200000e+01	0.000000e+00	0.000000e+00	0.000000	0.000000
50%	2017-10-14 00:00:00	1.190000e+02	1.000000e+00	1.000000e+00	0.000000	0.000000
75%	2018-01-21 00:00:00	1.328000e+03	1.000000e+00	9.000000e+00	0.000000	1.000000
max	2018-04-30 00:00:00	1.461329e+06	2.300000e+01	2.820000e+02	1.000000	2.000000
std	NaN	6.821058e+03	2.104095e+00	1.222015e+01	0.093593	0.494782

In [35]:
```python
# check if any column has null value in the dataset
dataframe.isnull().any()
```

Out[35]:
```
date_               False
user_id_maped       False
program_name        False
duration_seconds    False
program_class       False
season              False
episode             False
program_desc        False
program_genre       False
series_title         True
hd                   True
original_name       False
dtype: bool
```

Task 1

You are required to work on task one to study and HD flag for available dataset

In [45]:
```python
# make a copy of the dataframe for working on task 1
df=dataframe.copy()
grouped=df.copy()
grouped
```

Out[45]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0.0	0.0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0.0	1.0	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2017-04-22	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0.0	0.0	Extraterrestrial
1048571	2017-11-29	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0.0	0.0	Friends
1048572	2017-06-15	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0.0	0.0	Masters of Horror
1048573	2017-09-11	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0.0	1.0	Two Lovers and a Bear
1048574	2017-05-02	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0.0	1.0	American Psycho

1048575 rows × 12 columns

In [47]:
```python
# For series we concatenated the Session episode to differentiate between episodes

grouped.loc[grouped['program_class'] == 'SERIES/EPISODES', 'program_name'] = grouped['program_name']+'_SE'+grouped['season'].astype(str)+'_EP'+grouped['episode'].astype(str)

grouped.head(50)
```

Out[47]:

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0.0	0.0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0.0	1.0	Churchill
5	2017-08-19	20775	Beavis And Butt-Head Do America	3	MOVIE	0	0	Animation MovieBeavis And Butt-Head Do America	Animation	0.0	0.0	Beavis And Butt-Head Do America
6	2017-06-26	11374	The Mermaid Princess	33577	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
7	2018-03-29	6358	Coco	14	MOVIE	0	0	Animation MovieCoco (HD)	Animation	0.0	1.0	Coco
8	2018-01-27	11660	Kidnap	85	MOVIE	0	0	Action MovieKidnap (HD)	Action	0.0	1.0	Kidnap
9	2017-03-30	5155	The Accountant	42	MOVIE	0	0	Action MovieThe Accountant (HD)	Action	0.0	1.0	The Accountant
10	2017-04-10	30889	The Birth of a Nation	71	MOVIE	0	0	Biography MovieThe Birth of a Nation (HD)	Biography	0.0	1.0	The Birth of a Nation
11	2017-08-30	1035	The Amazing pider-Man	85	MOVIE	0	0	Action MovieThe Amazing Spider-Man	Action	0.0	0.0	The Amazing pider-Man
12	2018-02-02	12598	Baywatch	46	MOVIE	0	0	Comedy MovieBaywatch (HD)	Comedy	0.0	1.0	Baywatch
13	2017-09-24	7805	Alien: Covenant	90	MOVIE	0	0	Horror MovieAlien: Covenant (HD)	Horror	0.0	1.0	Alien: Covenant
14	2017-09-20	33846	Going in tyle	86	MOVIE	0	0	Comedy MovieGoing in Style (HD)	Comedy	0.0	1.0	Going in tyle
15	2018-01-20	12333	Kingsman: The Golden Circle	153	MOVIE	0	0	Action MovieKingsman: The Golden Circle (HD)	Action	0.0	1.0	Kingsman: The Golden Circle
16	2018-02-10	25343	Billy Elliot the Musical Live	2	MOVIE	0	0	Comedy MovieBilly Elliot the Musical Live (HD)	Comedy	0.0	1.0	Billy Elliot the Musical Live
17	2017-08-01	30971	Going in tyle	35	MOVIE	0	0	Comedy MovieGoing in Style (HD)	Comedy	0.0	1.0	Going in tyle
18	2017-06-25	26546	Moana	43	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
19	2017-06-23	20306	Moana	12	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
20	2017-08-13	16962	Pitch Perfect	28	MOVIE	0	0	Comedy MoviePitch Perfect (HD)	Comedy	0.0	1.0	Pitch Perfect
21	2017-03-26	16880	Surf's Up : WaveMania	84	MOVIE	0	0	Animation MovieSurf's Up 2: WaveMania (HD)	Animation	0.0	1.0	Surf's Up : WaveMania
22	2017-12-03	19694	Hell On Wheels The White pirit_SE2_EP7	2650	SERIES/EPISODES	2	7	Drama MovieThe White Spirit (HD)	Drama	0.0	0.0	Hell On Wheels
23	2017-05-15	22529	Sully	69	MOVIE	0	0	Biography MovieSully (HD)	Biography	0.0	1.0	Sully
24	2017-07-06	21917	Goat	12	MOVIE	0	0	Drama MovieGoat (HD)	Drama	0.0	1.0	Goat
25	2017-09-08	28952	Last Knights	47	MOVIE	0	0	Action MovieLast Knights	Action	0.0	0.0	Last Knights
26	2017-08-30	25404	Land Of The Dead	3553	MOVIE	0	0	Horror MovieLand Of The Dead (HD)	Horror	0.0	1.0	Land Of The Dead
27	2018-03-05	14270	Geostorm	2	MOVIE	0	0	Action MovieGeostorm (HD)	Action	0.0	1.0	Geostorm
28	2017-05-02	14249	Trolls	20	MOVIE	0	0	Animation MovieTrolls (HD)	Animation	0.0	1.0	Trolls
29	2017-09-29	21019	The Mermaid Princess	21	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
30	2018-01-16	17696	Inside	6	MOVIE	0	0	Horror MovieInside (HD)	Horror	0.0	1.0	Inside
31	2018-01-27	24323	Underworld: Awakening	127	MOVIE	0	0	Action MovieUnderworld: Awakening	Action	0.0	0.0	Underworld: Awakening
32	2018-01-30	13067	American niper	33	MOVIE	0	0	Action MovieAmerican Sniper	Action	0.0	0.0	American niper
33	2017-05-02	23833	Sully	88	MOVIE	0	0	Biography MovieSully (HD)	Biography	0.0	1.0	Sully
34	2018-02-16	15269	Game of Aces	34	MOVIE	0	0	Action MovieGame of Aces	Action	0.0	0.0	Game of Aces
35	2017-05-25	7092	Extraterrestrial	5	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0.0	0.0	Extraterrestrial
36	2017-09-02	3727	Lego DC uper Hero Girls: Brain Drain	42	MOVIE	0	0	Animation MovieLego DC Super Hero Girls: Brain...	Animation	0.0	0.0	Lego DC uper Hero Girls: Brain Drain
37	2017-06-28	25873	The Jetsons & WWE: Robo-WrestleMania!	8	MOVIE	0	0	Animation MovieThe Jetsons & WWE: Robo-Wrestle...	Animation	0.0	1.0	The Jetsons & WWE: Robo-WrestleMania!
38	2017-06-04	12863	The Mermaid Princess	53	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
39	2017-04-08	28626	Surf's Up : WaveMania	83	MOVIE	0	0	Animation MovieSurf's Up 2: WaveMania (HD)	Animation	0.0	1.0	Surf's Up : WaveMania
40	2017-08-01	1933	Lego cooby-Doo! Blowout Beach Bash	81	MOVIE	0	0	Animation MovieLego Scooby-Doo! Blowout Beach ...	Animation	0.0	1.0	Lego cooby-Doo! Blowout Beach Bash
41	2018-03-21	18348	Ferdinand	388356	MOVIE	0	0	Animation MovieFerdinand (HD)	Animation	0.0	1.0	Ferdinand
42	2018-04-08	5073	Geostorm	62	MOVIE	0	0	Action MovieGeostorm (HD)	Action	0.0	1.0	Geostorm
43	2017-12-08	23671	Baywatch	59	MOVIE	0	0	Comedy MovieBaywatch (HD)	Comedy	0.0	1.0	Baywatch
44	2018-03-20	19816	Geostorm	26	MOVIE	0	0	Action MovieGeostorm (HD)	Action	0.0	1.0	Geostorm
45	2018-03-12	24405	Monsters University	142	MOVIE	0	0	Animation MovieMonsters University	Animation	0.0	0.0	Monsters University
46	2017-10-25	934	War for the Planet of the Apes	14	MOVIE	0	0	Action MovieWar for the Planet of the Apes (HD)	Action	0.0	1.0	War for the Planet of the Apes
47	2017-04-22	23324	Surf's Up	42	MOVIE	0	0	Animation MovieSurf's Up	Animation	0.0	0.0	Surf's Up
48	2017-05-30	28687	Moana	61	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
49	2018-01-05	33459	Instant Death	39	MOVIE	0	0	Action MovieInstant Death (HD)	Action	0.0	1.0	Instant Death

In [49]:
```python
grouped= grouped.groupby(['program_name','program_class'])\
.agg({'user_id_maped': [('co1', 'nunique'),('co2', 'count')],\
     'duration_seconds': [('co3', 'sum')] }).reset_index()
grouped
```

Out[49]:

	program_name	program_class	user_id_maped		duration_seconds
			co1	co2	co3
0	#FollowFriday	MOVIE	316	510	191769
1	10 Days in a Madhouse	MOVIE	301	553	356806
2	100 treets	MOVIE	704	1851	319000
3	101 Dalmatians	MOVIE	85	118	30002
4	102 Dalmatians	MOVIE	130	172	32235
...	...	...	...	...	...
11362	iCarly iWant More Viewers_SE1_EP3	SERIES/EPISODES	76	217	172733
11363	iCarly iWill Date Freddie_SE1_EP8	SERIES/EPISODES	61	183	295301
11364	iCarly_SE1_EP0	SERIES/EPISODES	73	133	14220
11365	xXx	MOVIE	58	94	60465
11366	xXx: Return of Xander Cage	MOVIE	402	853	938948

11367 rows × 5 columns

In [51]:
```python
grouped.columns = ['program_name','program_class','No of Users who Watched', 'No of watches', 'Total watch time in seconds']
grouped
```

Out[51]:

	program_name	program_class	No of Users who Watched	No of watches	Total watch time in seconds
0	#FollowFriday	MOVIE	316	510	191769
1	10 Days in a Madhouse	MOVIE	301	553	356806
2	100 treets	MOVIE	704	1851	319000
3	101 Dalmatians	MOVIE	85	118	30002
4	102 Dalmatians	MOVIE	130	172	32235
...	...	...	...	...	...
11362	iCarly iWant More Viewers_SE1_EP3	SERIES/EPISODES	76	217	172733
11363	iCarly iWill Date Freddie_SE1_EP8	SERIES/EPISODES	61	183	295301
11364	iCarly_SE1_EP0	SERIES/EPISODES	73	133	14220
11365	xXx	MOVIE	58	94	60465
11366	xXx: Return of Xander Cage	MOVIE	402	853	938948

11367 rows × 5 columns

In [53]:
```python
grouped['Total watch time in houres']=grouped['Total watch time in seconds']/3600
grouped
```

Out[53]:

	program_name	program_class	No of Users who Watched	No of watches	Total watch time in seconds	Total watch time in houres
0	#FollowFriday	MOVIE	316	510	191769	53.269167
1	10 Days in a Madhouse	MOVIE	301	553	356806	99.112778
2	100 treets	MOVIE	704	1851	319000	88.611111
3	101 Dalmatians	MOVIE	85	118	30002	8.333889
4	102 Dalmatians	MOVIE	130	172	32235	8.954167
...	...	...	...	...	...	...
11362	iCarly iWant More Viewers_SE1_EP3	SERIES/EPISODES	76	217	172733	47.981389
11363	iCarly iWill Date Freddie_SE1_EP8	SERIES/EPISODES	61	183	295301	82.028056
11364	iCarly_SE1_EP0	SERIES/EPISODES	73	133	14220	3.950000
11365	xXx	MOVIE	58	94	60465	16.795833
11366	xXx: Return of Xander Cage	MOVIE	402	853	938948	260.818889

11367 rows × 6 columns

In [55]:
```python
grouped = grouped.drop(columns=['Total watch time in seconds'])
grouped
```

Out[55]:

	program_name	program_class	No of Users who Watched	No of watches	Total watch time in houres
0	#FollowFriday	MOVIE	316	510	53.269167
1	10 Days in a Madhouse	MOVIE	301	553	99.112778
2	100 treets	MOVIE	704	1851	88.611111
3	101 Dalmatians	MOVIE	85	118	8.333889
4	102 Dalmatians	MOVIE	130	172	8.954167
...	...	...	...	...	...
11362	iCarly iWant More Viewers_SE1_EP3	SERIES/EPISODES	76	217	47.981389
11363	iCarly iWill Date Freddie_SE1_EP8	SERIES/EPISODES	61	183	82.028056
11364	iCarly_SE1_EP0	SERIES/EPISODES	73	133	3.950000
11365	xXx	MOVIE	58	94	16.795833
11366	xXx: Return of Xander Cage	MOVIE	402	853	260.818889

11367 rows × 5 columns

In [57]:
```python
grouped = grouped.sort_values(by=['Total watch time in houres', 'No of watches','No of Users who Watched'], ascending=False).reset_index(drop=True)
grouped
```

Out[57]:

	program_name	program_class	No of Users who Watched	No of watches	Total watch time in houres
0	The Boss Baby	MOVIE	3389	24047	2961.350833
1	The Amazing pider-Man	MOVIE	1011	2877	1966.119167
2	The Expendables	MOVIE	853	2119	1961.159444
3	Moana	MOVIE	2173	8081	1706.176944
4	Trolls	MOVIE	2613	13793	1601.023056
...	...	...	...	...	...
11362	Mad Love	MOVIE	1	1	0.001111
11363	Al Akhwa A'adaa Ep30_SE1_EP30	SERIES/EPISODES	1	1	0.000833
11364	Mumuhug_SE5_EP10	SERIES/EPISODES	1	1	0.000833
11365	Thomas And Friends Ep_SE9_EP5	SERIES/EPISODES	1	1	0.000833
11366	The Chair The Ramp Up_SE1_EP4	SERIES/EPISODES	1	1	0.000556

11367 rows × 5 columns

In [59]:
```python
# show the result
grouped.head(35)
```

Out[59]:

	program_name	program_class	No of Users who Watched	No of watches	Total watch time in houres
0	The Boss Baby	MOVIE	3389	24047	2961.350833
1	The Amazing pider-Man	MOVIE	1011	2877	1966.119167
2	The Expendables	MOVIE	853	2119	1961.159444
3	Moana	MOVIE	2173	8081	1706.176944
4	Trolls	MOVIE	2613	13793	1601.023056
5	Bean	MOVIE	949	3617	1423.955000
6	The murfs	MOVIE	867	3132	1342.141111
7	Hotel Transylvania	MOVIE	491	1947	1096.533611
8	Cloudy With a Chance of Meatballs	MOVIE	683	2076	948.674722
9	The Man With The Iron Fists	MOVIE	707	2505	859.626389
10	Salt	MOVIE	563	1082	767.392778
11	Unbroken	MOVIE	625	1429	763.078333
12	ParaNorman	MOVIE	614	1746	747.065556
13	Youm Maloosh Lazma	MOVIE	1131	2278	718.109722
14	Ferdinand	MOVIE	1278	6817	714.223056
15	White Chicks	MOVIE	307	916	711.840833
16	Jurassic Park	MOVIE	504	1192	693.394444
17	The November Man	MOVIE	494	1219	679.492222
18	Total Recall	MOVIE	587	1108	661.820000
19	Robin Hood	MOVIE	588	1209	643.935000
20	Public Enemies	MOVIE	368	716	634.035000
21	Daddy Day Camp	MOVIE	263	647	625.338333
22	Oblivion	MOVIE	790	1678	609.391111
23	Blitz	MOVIE	562	1200	570.521944
24	War for the Planet of the Apes	MOVIE	879	2028	567.597778
25	Inside Man	MOVIE	532	1567	560.386111
26	Bad Boys	MOVIE	438	871	559.277500
27	Easy A	MOVIE	513	990	557.068611
28	Battleship	MOVIE	634	1324	552.857222
29	Baywatch	MOVIE	2062	7436	548.995556
30	Police tory	MOVIE	409	737	520.077222
31	Blindspot Erase Weary Youth_SE1_EP13	SERIES/EPISODES	125	197	516.015556
32	47 Ronin	MOVIE	521	1701	515.014722
33	Alien: Covenant	MOVIE	1520	4260	507.180000
34	The Mermaid Princess	MOVIE	2182	7525	503.111111

In [ ]:
```python
```

In [61]:
```python
# we import Visualization libraries
# you can ignore and use any other graphing libraries
import matplotlib.pyplot as plt # a comprehensive library for creating static, animated, and interactive visualizations
import plotly #a graphing library makes interactive, publication-quality graphs. Examples of how to make line plots, scatter plots, area charts, bar charts, error bars, box plots, histograms, heatmaps, subplots, multiple-axes, polar
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

In [71]:
```python
# plot top 10 Programs
fig = px.pie(grouped.head(10), values='Total watch time in houres', names='program_name',\
            hover_data=['program_class'],title='top 10 programs in total watch time in houres')
fig.show()
```

In [103…
```python
# Here we try to study the customer experience against Program class
grouped=df.copy()
grouped
```

Out[103…

	date_	user_id_maped	program_name	duration_seconds	program_class	season	episode	program_desc	program_genre	series_title	hd	original_name
0	2017-05-27	26138	100 treets	40	MOVIE	0	0	Drama Movie100 Streets	Drama	0.0	0.0	100 treets
1	2017-05-21	7946	Moana	17	MOVIE	0	0	Animation MovieMoana (HD)	Animation	0.0	1.0	Moana
2	2017-08-10	7418	The Mermaid Princess	8	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
3	2017-07-26	19307	The Mermaid Princess	76	MOVIE	0	0	Animation MovieThe Mermaid Princess (HD)	Animation	0.0	1.0	The Mermaid Princess
4	2017-07-07	15860	Churchill	87	MOVIE	0	0	Biography MovieChurchill (HD)	Biography	0.0	1.0	Churchill
...	...	...	...	...	...	...	...	...	...	...	...	...
1048570	2017-04-22	23716	Extraterrestrial	4865	MOVIE	0	0	Horror MovieExtraterrestrial (HD)	Horror	0.0	0.0	Extraterrestrial
1048571	2017-11-29	1776	Friends The One With The Holiday Armadillo	1261	SERIES/EPISODES	7	10	Comedy MovieThe One With The Holiday Armadillo...	Comedy	0.0	0.0	Friends
1048572	2017-06-15	1918	Masters of Horror Dreams In The Witch House	6	SERIES/EPISODES	1	2	Horror MovieDreams In The Witch House	Horror	0.0	0.0	Masters of Horror
1048573	2017-09-11	25737	Two Lovers and a Bear	72	MOVIE	0	0	Drama MovieTwo Lovers and a Bear (HD)	Drama	0.0	1.0	Two Lovers and a Bear
1048574	2017-05-02	19844	American Psycho	2685	MOVIE	0	0	Drama MovieAmerican Psycho (HD)	Drama	0.0	1.0	American Psycho

1048575 rows × 12 columns

In [105…
```python
grouped = grouped.groupby('program_class')\
.agg({'user_id_maped': [('co1', 'nunique'),('co2', 'count')],\
     'duration_seconds': [('co3', 'sum')] }).reset_index()
grouped
```

Out[105…

	program_class	user_id_maped		duration_seconds
		co1	co2	co3
0	MOVIE	11355	488401	372398924
1	SERIES/EPISODES	3901	560174	918352035

In [107…
```python
grouped.columns = ['program_class','No of Users who Watched', 'No of watches', 'Total watch time in seconds']

grouped
```

Out[107…

	program_class	No of Users who Watched	No of watches	Total watch time in seconds
0	MOVIE	11355	488401	372398924
1	SERIES/EPISODES	3901	560174	918352035

In [109…
```python
grouped['Total watch time in houres']=grouped['Total watch time in seconds']/3600
grouped
```

Out[109…

	program_class	No of Users who Watched	No of watches	Total watch time in seconds	Total watch time in houres
0	MOVIE	11355	488401	372398924	103444.145556
1	SERIES/EPISODES	3901	560174	918352035	255097.787500

In [111…
```python
grouped = grouped.drop(columns=['Total watch time in seconds'])
grouped
```

Out[111…

	program_class	No of Users who Watched	No of watches	Total watch time in houres
0	MOVIE	11355	488401	103444.145556
1	SERIES/EPISODES	3901	560174	255097.787500

In [113…
```python
grouped = grouped.sort_values(by=['Total watch time in houres', 'No of watches','No of Users who Watched'], ascending=False).reset_index(drop=True)
grouped
```

Out[113…

	program_class	No of Users who Watched	No of watches	Total watch time in houres
0	SERIES/EPISODES	3901	560174	255097.787500
1	MOVIE	11355	488401	103444.145556

In [ ]:
```python
```

In [120…
```python
# show the result
grouped.head()
```

Out[120…

	program_class	No of Users who Watched	No of watches	Total watch time in houres
0	SERIES/EPISODES	3901	560174	255097.787500
1	MOVIE	11355	488401	103444.145556

In [122…
```python
# plot the total watch time against total number of users and report your findings
fig = px.pie(grouped, values='Total watch time in houres', names='program_class',\
            hover_data=['program_class'],title='Total duration spent by program_class')



fig2 = px.pie(grouped, values='No of Users who Watched', names='program_class',\
            hover_data=['program_class'],title='Total Users watching by program_class')



fig.update_traces(sort=False)
fig2.update_traces(sort=False)
fig.show()
fig2.show()
```

Task

In [ ]:
```python
"""
TODO study the relation and user's behaviour Against HD flag, feel free to include any comparison
"""
```

In [124…
```python
grouped_hd = df.copy()


grouped_hd = grouped_hd.groupby('hd')\
.agg({
    'user_id_maped': [('co1', 'nunique'), ('co2', 'count')],
    'duration_seconds': [('co3', 'sum')]
}).reset_index()


grouped_hd.columns = ['HD Flag', 'No of Users who Watched', 'No of watches', 'Total watch time in seconds']


grouped_hd['Total watch time in hours'] = grouped_hd['Total watch time in seconds'] / 3600


grouped_hd = grouped_hd.drop(columns=['Total watch time in seconds'])


grouped_hd = grouped_hd.sort_values(by=['Total watch time in hours', 'No of watches', 'No of Users who Watched'], ascending=False).reset_index(drop=True)


grouped_hd
```

Out[124…

	HD Flag	No of Users who Watched	No of watches	Total watch time in hours
0	0.0	6714	519407	225863.346389
1	1.0	10948	385415	86143.602500
2	2.0	57	118	47.871667

In [126…
```python
fig = px.bar(grouped_hd,
             x='HD Flag',
             y=['Total watch time in hours', 'No of watches'],
             title='User Behavior Against HD Flag',
             labels={'value': 'Count', 'HD Flag': 'HD Flag'},
             barmode='group')

fig.show()
```

In [128…
```python
fig = px.pie(grouped_hd, values='Total watch time in hours', names='HD Flag', title='Percentage of Total Watch Time by HD Flag')
fig.show()
```

In [ ]:
```python
```

In [ ]:
```python
```